

Education

Massachusetts Institute of Technology

Sep. 2025 - Present

PH.D. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE, EXPECTED JUN. 2030

- GPA: **5.0/5.0**
- Relevant Coursework: Deep Learning (**A+**), AI for Protein Biology (A), Advances in Computer Vision (A)

Tsinghua University

Sep. 2021 - Jul. 2025

B.ENG. IN COMPUTER SCIENCE AND ENGINEERING

- GPA: **3.94/4.0 (19/178)**

Technical Skills

Languages & Frameworks

Python, C/C++, PyTorch, Hugging Face Transformers, DeepSpeed

Infrastructure

Distributed model training (100B-scale), CUDA, Slurm, Docker, Git

Publications

- Bo Chen*, **Zhilei Bei***, Xingyi Cheng, Pan Li, Jie Tang, Le Song. *MSAGPT: Neural Prompting for Protein Structure Prediction via MSA Generative Pre-Training*. *NeurIPS* (2024). * Equal contribution.
- Bo Chen*, Xingyi Cheng*, Pan Li, Yangli-ao Geng, Jing Gong, Shen Li, **Zhilei Bei**, et al. *xTrimoPGLM: Unified 100B-Scale Pre-Trained Transformer for Deciphering the Language of Proteins*. *Nature Methods* (2025). * Equal contribution.
- Ajinkya Deshpande, **Zhilei Bei**, Jian Ma, Spencer Krieger. *MIMYR: Generative Modeling of Missing Tissue in Spatial Transcriptomics*. *Gen2 Workshop at ICLR* (2026).
- Boyan Wang, Yangliao Geng, Xingyi Cheng, Bo Chen, **Zhilei Bei**, et al. *ProtGO: Universal Protein Function Prediction Utilizing Multi-Modal Gene Ontology Knowledge*. *Bioinformatics* (2025).

Research Experience

Massachusetts Institute of Technology

Ph.D. Researcher, Hwang Lab

Sep. 2025 - Present

- **Interpretable Learning of Bacterial Regulatory Networks from Genome Sequence** (Hwang Lab, 2026–)
 - Developing an interpretable machine learning framework to predict gene expression and infer regulatory relationships from microbial genome sequence.
 - Recovering known regulatory interactions without direct interaction labels and outperforming established bioinformatics baselines on [metric/result].
- **Context-aware Protein Subcellular Localization Prediction** (rotation: Kaiming He, Le Cong)
 - Fused protein LM embeddings with landmark-staining images and transcriptomes ([[N]] proteins, [[M]] cell lines).
 - Improved [[macro-F1]] [[X%]] on unseen cell lines over a sequence-only baseline, isolating context's contribution.
- **Protein Metal Binding Site Prediction for Metalloprotein Discovery** (rotation: Yunha Hwang)
 - Built a sequence-based metal-binding site predictor for metalloprotein screening ([[N]] sites, [[M]] metals).
 - Swapped metal-specific heads for a joint protein-metal embedding space: zero-shot transfer ([[+X]] AUPRC).
- **Few-shot Conditional Generation for Protein Design** (rotation: Alex Rives)
 - Built a masked diffusion language model generating proteins from [[K]] few-shot family examples at inference.
 - Added classifier-free guidance to steer sampling, gaining [[X%]] [[predicted fitness]] over unguided sampling.

Zhipu AI

Student Researcher

Feb. 2025 - Jun. 2025

- **Reinforcement Learning for Self-Improvement in Machine Learning Agents**

- Designed agent scaffolding for long-horizon ML engineering, with a memory module accumulating lessons.
- Applied RL over single-step decisions to train on high-information trajectories: [[medal rate]] [[X]]→[[Y]].

Carnegie Mellon University

Research Assistant, Advisor: Prof. Jian Ma

Jun. 2024 - Sep. 2025

- **Predicting Missing Tissue Regions in Spatial Transcriptomics**

- Paired a VAE over expression profiles with a ViT that inpaints spatial transcriptomics for missing tissue regions.
- Reduced [[imputation error]] [[X%]] over [[baseline]] on held-out regions; ICLR 2026 Gen2 Workshop.

- **Protein Language Foundation Models** (xTrimopGLM, 100B parameters; *Nature Methods* 2025)
 - Ran scaling studies to 100B parameters, showing understanding and generation objectives scale differently.
 - Raised [[generation metric]] from [[a]] to [[b]] via supervised fine-tuning and reinforcement self-training.
- **Protein Multiple Sequence Alignment (MSA) Generation** (MSAGPT, NeurIPS 2024)
 - Built a generative model synthesizing virtual MSAs for low-homology proteins ([[+X]] AlphaFold2 TM-score).
 - Used AlphaFold2 structural feedback as the RL reward, further improving TM-score by [[Y]].

- **Pocket-aware Ligand Generation with Diffusion Models**
 - Guided pocket-conditioned diffusion with a GNN affinity predictor and unpaired-data augmentation: +13% [[Vina]].

- **Disease Target and Drug Discovery Using Community Detection on Knowledge Graphs**
 - Ranked therapeutic targets and drugs for breast cancer from knowledge-graph embeddings + community detection.

- **Blocking the Binding of Bacterial Lipid Droplets to Phage DNA**
 - Identified a blocker of bacterial lipid droplet–phage DNA binding *in vitro* (confocal microscopy, blotting).

Honors & Awards

2025 **Sunlin (1966) and Priscilla Chou Fellowship**, MIT EECS Graduate Student Fellowship

2024 **Spark Scientific and Technological Innovation Fellowship**, Top 1% of Tsinghua University

2024 **Zhong Shimo Scholarship**, Top 1% of Computer Science Department, Tsinghua University

2024 **National Scholarship**, Top 3% of Tsinghua University

2023 **Comprehensive Scholarship**, Top 10% of Tsinghua University

2022 **Comprehensive Scholarship**, Top 10% of Tsinghua University

2020 **Finalist**, Regeneron International Science and Engineering Fair (ISEF)